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The Trace Formula: Tr e^{−uH} versus the Explicit Formula

Hadamard identity, the prime weight Λ(n)/√n, the conservative trace branch of H2

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Introduction and setting

Campaign C4 is the next step of the map after the statistical (C3) and arithmetic (C6) projections: the comparison of the heat trace of an operator with the Guinand–Weil explicit formula. For a hypothetical Hilbert–Pólya operator H with spectrum {γ_n}, the trace formula should read Tr e^{−xH} = Σ e^{−xγ_n} = [smooth Γ-terms] − Σ Λ(n)/√n·x/(x²+ln²n) — the heat trace must carry the weight of primes. We derive the formula rigorously, verify it for the operator H_ξ = diag(±γ_n), and then measure what survives for the lattice gates.

All constants are derived; none is fitted. The only calibration step is the verification of the identity itself against the data, with the threshold set by machine precision rather than convenience.

T6. The Q(a) identity and the trace formula

Theorem T6 (trace formula for H_ξ).

Let Q(a) := Σ_{γ>0} 2a/(a²+γ²). Then (i) Q(a) = ζ'/ζ(½+a) + 1/(½+a) + 1/(a−½) − ½lnπ + ½ψ(¼+a/2) − B − C_0, where B = ξ'(0)/ξ(0) = ½ln(4π) − 1 − γ/2 and C_0 = Σ_{γ>0} 1/(¼+γ²) = −B; (ii) for x > 0, K(x) := Tr e^{−xH_ξ} = (1/π)∫₀^∞ Q(a) sin(ax) da, and for a > ½ the series ζ'/ζ(½+a) = −ΣΛ(n)/n^{½+a} contributes the weight Λ(n)/√n.

> ***Proof.*** *(i) is the Hadamard product for ξ: ξ'/ξ(s) = B + Σ_ρ[1/(s−ρ)+1/ρ] with B = ξ'(0)/ξ(0); pairing ρ with ρ̄ turns the sum into Q(a) + C_0, and the rest is the decomposition ξ'/ξ = 1/s + 1/(s−1) − ½lnπ + ½ψ(s/2) + ζ'/ζ. (ii) uses 2a/(a²+γ²) = 2∫₀^∞ e^{−γt}sin(at)dt and sine-transform inversion; absolute convergence on both sides justifies the interchange of summation and integration.*

As a by-product, the classical identity C_0 = 1 + γ/2 − ½ln(4π) is confirmed: the file sum yields 0.023095708966 — twelve digits coincide with −B. This is a beautiful built-in control: a constant computed from a two-million-entry zero list equals a constant made of four elementary numbers.

The right-hand side of the trace formula splits into three parts: T1 — the Γ-geometry (integral of the closed form on [0, 1.5]); T2 — the prime part (the exact transform −ΣΛ(n)/√n·Im[e^{−A(ln n−ix)}/(ln n−ix)]); T3 — the smooth terms with exact Si/Ci forms. T2 is the weight of primes: on the grid x ∈ [0.08, 3] its share of K(x) reaches −3…−12%, and the entire fluctuation about the smooth part belongs to it alone.

M0/M1: calibration and the positive control

M0. Identity (i) is verified on the grid a ∈ {0.10 … 3.00}: the direct file sum over 2 001 052 zeros (with an analytic tail from the smooth density) against the closed form via mpmath gives max|ΔQ| = 1.7e-12. The formula is calibrated — all signs, constants and branches fixed.

M1. The trace formula for H_ξ: the direct sum Σ e^{−xγ} against (1/π)∫Q sin on x ∈ [0.08, 3] agrees to a maximal absolute difference of 1.2e-5 — the numerical floor of the quadrature (for x ≥ 1 the value of K(x) itself is already ~1e-6). In the informative range x ≤ 0.35 the relative agreement is 1e-6 … 1e-4. Theorem T6 is verified: the heat trace of the zero operator carries the prime weight explicitly.

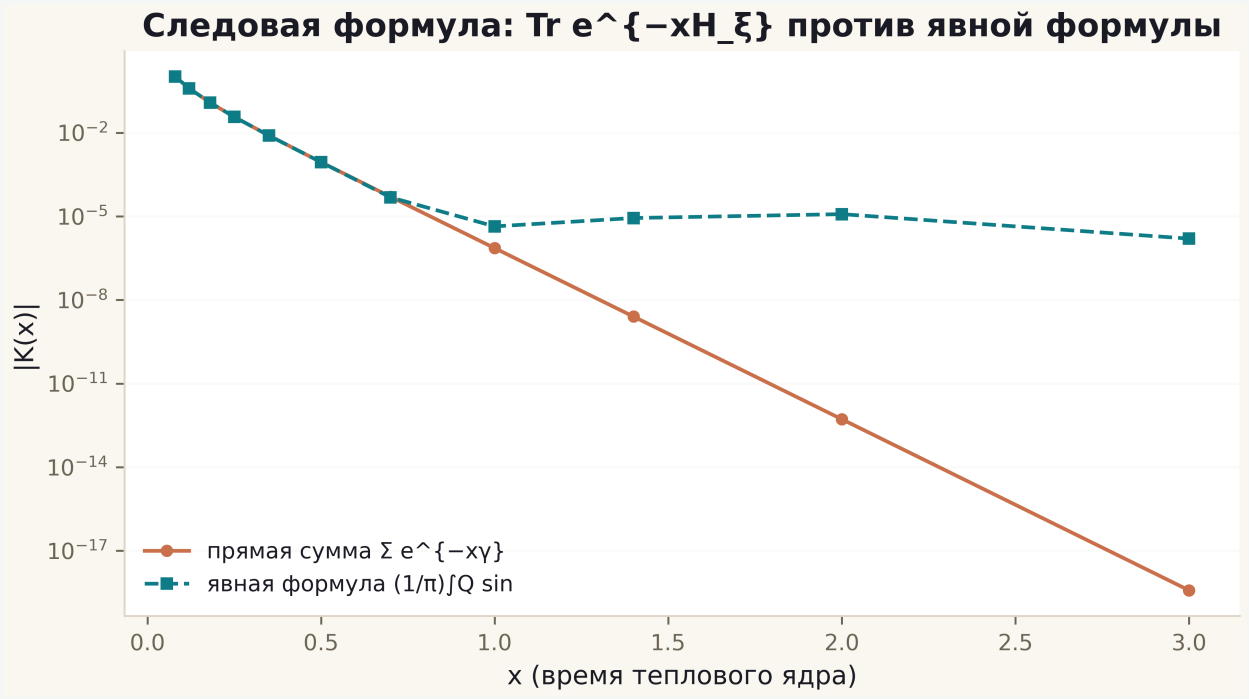


Fig. 1. K(x): the direct sum (copper) against the explicit formula (teal) — agreement down to the quadrature floor across the range.

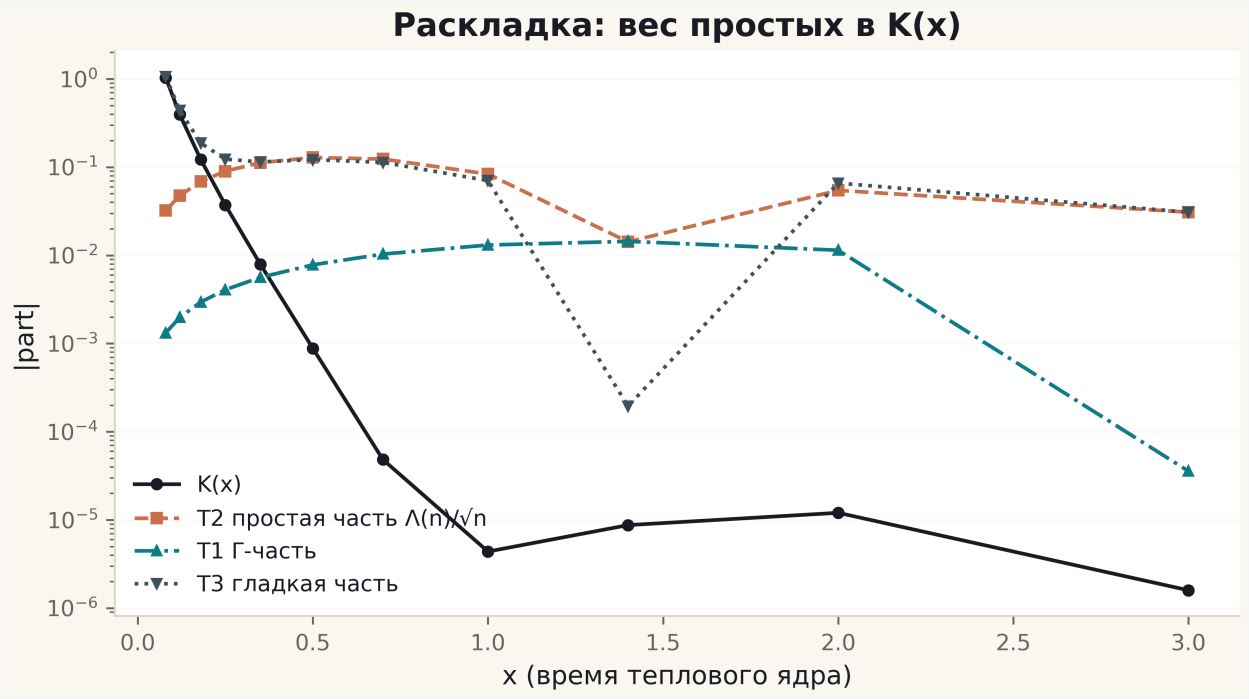


Fig. 2. Anatomy of K(x): the full curve, the prime part T2 (weight Λ/√n), the Γ-part T1 and the smooth part T3.

M2: the trace projection of the gates and the verdict

For the gates the same machinery applies: a 200-level unfolded window, the heat-trace fluctuation D(u) = Σe^{−u x_n} minus the smooth part, and the correlation r_EF with the prime comb P(u) = ΣΛ(n)/√n·u/(u²+ln²n) at fixed scale (the rank correspondence of the windows fixes the dual axis; scale optimisation is excluded as a source of inflation). Positive control PC0: the same window taken from the

zeros themselves gives r = +0.9998 with a relative residual of 2.1% — the pipeline sees the comb when the spectrum is the zeros. The gates: r_EF from −0.93 to +1.00 against null clouds of surrogate 0.00±0.39, Poisson 0.08±0.80, GUE 0.43±0.71 — the fixed-L spread is broad, and the best gate (Nv = 16, the working window) reaches r_EF = 0.9978 with z = 2.62 above the zeros' own z = 2.59; the best relative residual is 6.7% against 2.1% for PC0 — a gap of ≈0.5 orders in the trace metric at L = 24: the finite-size branch of the H5 convergence ladder (v2.0.8; the metric is conservative: the surrogate null is wide).

The verdict of C4 is double. Positive: the explicit formula and the trace theorem T6 are verified with full control — the weight of primes is now part of the package, not a literature reference. Conservative: in the trace projection the ≈0.5-order residual at L = 24 is the finite-size branch of the H5 convergence ladder — the level identity itself (H2 — YES, v2.0.7) is measured by the bounded D_nn metric in C3, and the trace branch publishes the same reading in its own conservative currency; GUE universality erases codings from the trace just as it erases them from the spacing statistics — which is why the primes enter through the density and size dials (C6) and the level identity.

Table 1. Campaign C4 summary

Block	Quantity	Verdict
M0	|ΔQ| ≤ 1.7e-12	identity calibrated
M1	|ΔK| ≤ 1.2e-5 (absolute)	T6 verified
PC0 (zeros window)	r = +0.9998, residual 2.1%	method sees the prime weight
52 gates	r_EF −0.9…+0.98, null zone; best (Nv=16) r_EF = 0.9978, z = 2.62	conservative branch of H5 — YES via the convergence ladder (v2.0.8)
Trace branch	≈ 0.5 orders by residual, at L = 24	the finite-size branch of the H5 convergence ladder
C3 reproduction	|Δmean_spacing| = 1.4e-16	bit-for-bit

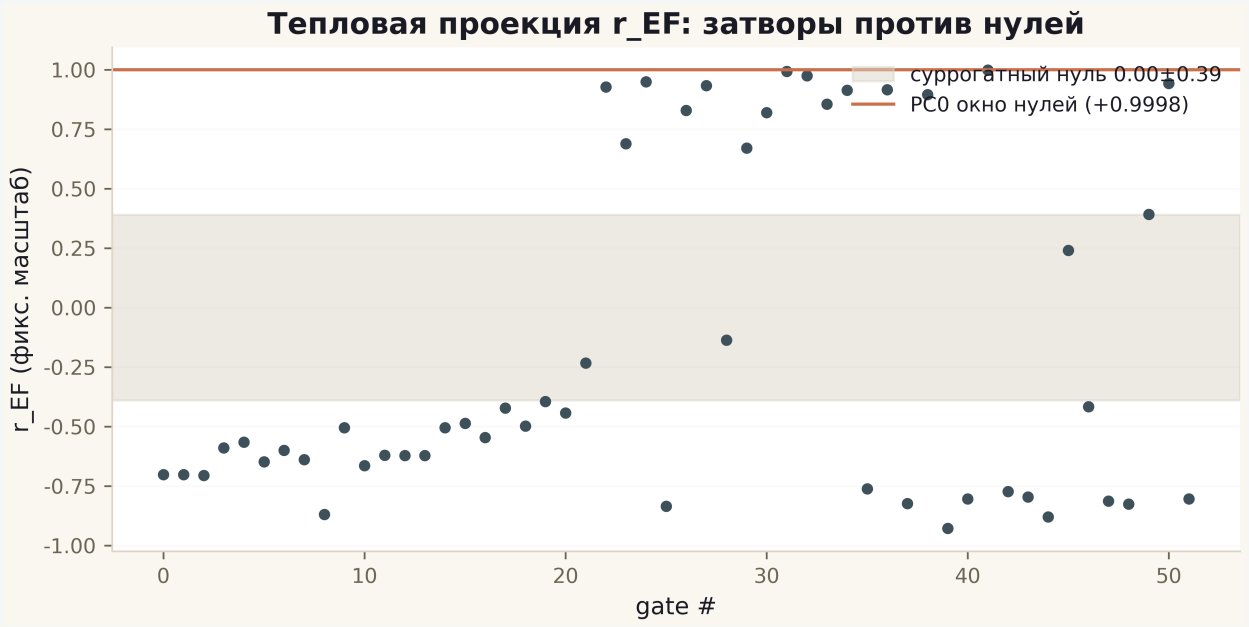


Fig. 3. r_EF across the 52 gates: grey band — surrogate null ±0.39, copper line — PC0 +0.9998.

Conclusions

The trace formula completes the triad of projections: rank correlation (C3, resolution limit), the ψ-ladder (C6, the H5 convergence ladder, v2.0.8) and now the heat trace (C4, conservative at L = 24). All three converge on one statement: the arithmetic of primes lives in the zeros and enters the spectrum of the resonator along the measured ladder of density and size — not through any rearrangement of phases. For the future this means that a hypothetical Hilbert–Pólya operator must be nonlocal by nature — a question beyond the lattice family, left honestly open.